Unit 12, Lesson 5: Factoring Common Monomials Factors

Benchmark

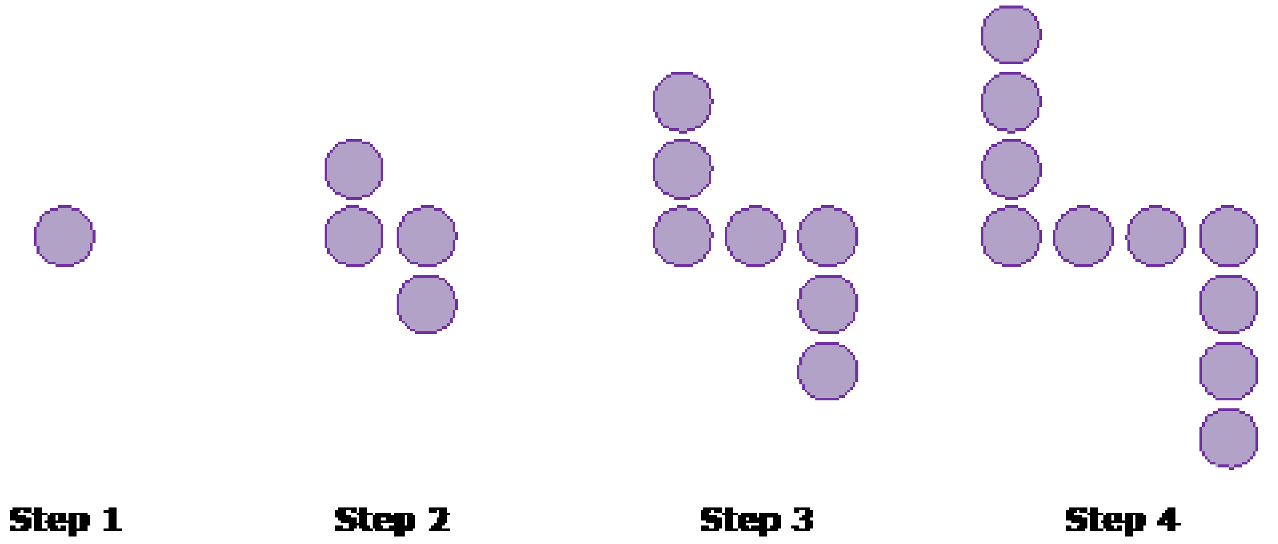
MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

Learning Target

- ☐ I can rewrite an algebraic expression with two terms that have a common monomial factor using the distributive property.

12.5.1 Warm-Up: Visual Patterns

1. The first four steps in a pattern are shown.



a. Describe how the pattern is changing in each step.

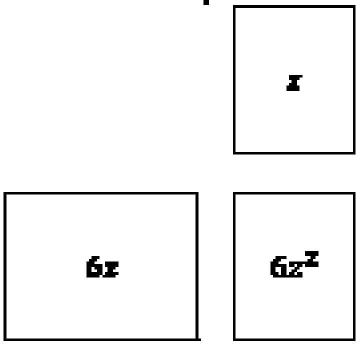
b. How many purple circles will be in step 5?

c. Draw step 10 of the pattern.

12.5.2 Exploration: Finding Common Factors of Monomials

1. Consider the monomial $6z^2$.

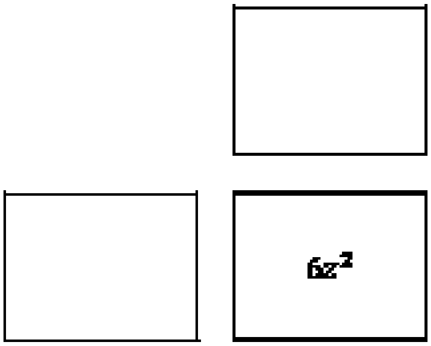
   a. The area model below shows two possible factors of $6z^2$.



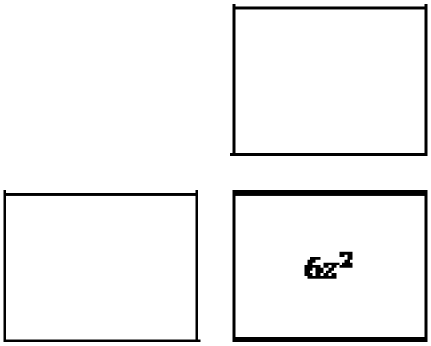
   b. Complete the equation by filling in the blanks with the factors from the area model.

$6z^2 =$ _____ · _____

   c. Work with your partner to identify two other possible pairs of factors for the monomial $6z^2$.



$6z^2 =$ _____ · _____



$6z^2 =$ _____ · _____

   d. Explain what strategy you used to identify factor pairs that multiply to $6z^2$.

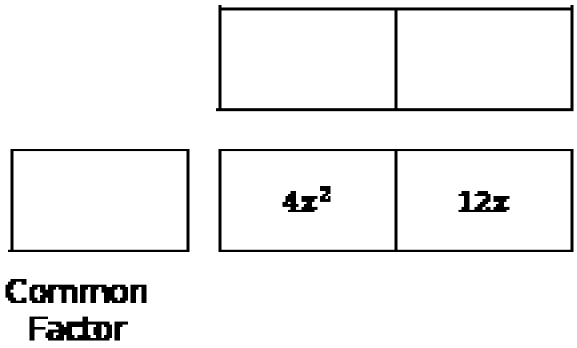
2. Consider the expression $4x^2 + 12x$.

a. Work with your partner to determine which of the terms shown are common factors of $4x^2$ and $12x$.

☐ 4	☐ x^2	☐ $4x$	☐ $2x$
☐ $6x$	☐ 2	☐ 6	☐ x

b. Explain how you made your choices.

c. Choose one of the common factors from Part A. Work with your partner to complete the area model using the common factor.



d. Rewrite the expression $4x^2 + 12x$ in factored form using the factors determined in Part C.

_____(_____ + _____)

The expression $4x^2 + 12x$ can be rewritten using other common factors.

e. Work with your partner to find three other equivalent expressions using different common factors of $4x^2$ and $12x$.

_____(_____ + _____)

_____(_____ + _____)

_____(_____ + _____)

12.5.3 Bring It Together: Finding Common Factors of Monomials

Similar to whole numbers, common factors of monomials can be found by considering the factors each term shares.

1. Work with your partner to complete the table. The first row has been completed.

Monomial Terms	Which factor(s) do the coefficients share?	Which variable factor(s) do the terms share?	Common Factors of the Monomials
$15w^2$ and $30w$	3, 5, 10	w	$3w$, $5w$, $10w$
$4c$ and $16c^2$			
$10g^2h^2$ and $20gh^2$			

Common factors of monomial terms can be used to rewrite sums of two algebraic expressions in an equivalent factored form.

2. Consider the expression $-12qr^2 + 20q^2r^2$.

a. Complete the area models in the table to find two equivalent expressions in factored form.

Area Model	Equivalent Expression
$4q$	$-12qr^2$	$20q^2r^2$	
$-2qr^2$	$-12qr^2$	$20q^2r^2$	

b. Explain how the first expression would change if $-4q$ had been used as the common factor instead of $4q$.

12.5.4 Guided Practice: Rewriting Expressions Using a Common Monomial Factor

1. Use the area model to find an equivalent expression to $32ab^2 + 8a^2b^2$ in factored form.

Area Model	Equivalent Expression
$32ab^2$ | $8a^2b^2$	____(____ + ____)

2. Consider the expression $14jk^2 - 35j^2k^2$.

   a. Complete the area models in the table to find two equivalent expressions in factored form.

Area Model	Equivalent Expression
$14jk^2$ | $-35j^2k^2$	
$14jk^2$ | $-35j^2k^2$	

   b. Verify that the expressions are equivalent to $14jk^2 - 35j^2k^2$ using the distributive property.

12.5.5 Collaboration: Error Analysis

One student's work rewriting sums of algebraic expressions using common factors is shown in the table. Work with your partner to analyze the student's work and complete the table.

Original Problem and Student Work	Are there any errors?	If there was an error, circle the error in the original work and write a correction.
1. Write an equivalent expression to $6f + 14f^2$. 3f | 7f 2f | 6f | 14f² $2f(3f + 7f)$	o Yes o No	

Original Problem and Student Work	Are there any errors?	If there was an error, circle the error in the original work and write a correction.
2. Write an equivalent expression to $3m^2n + 15m^2n^2$. $\begin{array}{c\|c\|c} & 1 & 5n \\ \hline 3m^2n & 3m^2n & 15m^2n^2 \end{array}$ $3m^2n(1 + 5n)$	o Yes o No	
3. Write an equivalent expression to $-25jk^2 + 45jk$. $\begin{array}{c\|c\|c} & -5k & 9jk \\ \hline 5jk & -25jk^2 & 45jk \end{array}$ $5jk(-5k + 9jk)$	o Yes o No	

12.5.6 Wrap-Up: Growth Mindset

Complete each statement based on today's lesson.

1. Today I am still unsure about ...

2. I will try to build my confidence by ...